QuizNova

Java-Based Online Quiz and Performance Management System

Student name:

Roll number:

Team:

Problem and project scope

Java practice needs a reliable timer, useful feedback and controlled publishing.

Participant

Register, take approved quizzes, review answers and track performance.

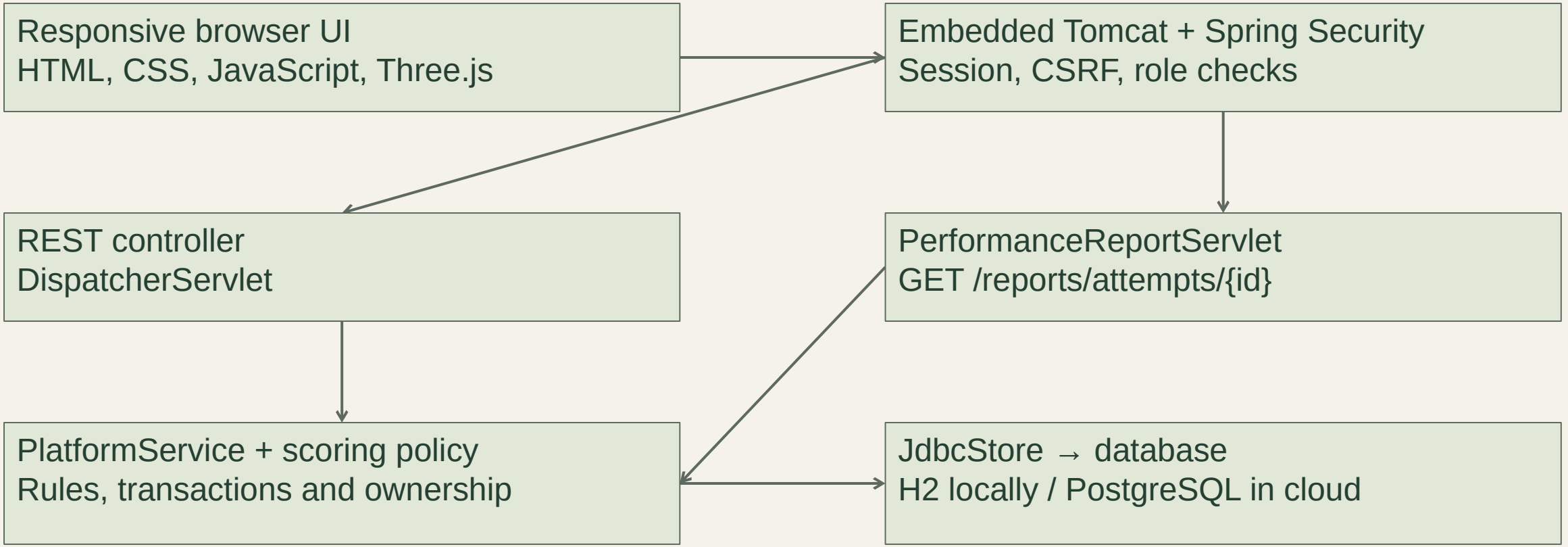
Quiz creator

Prepare questions and explanations, submit content and review results.

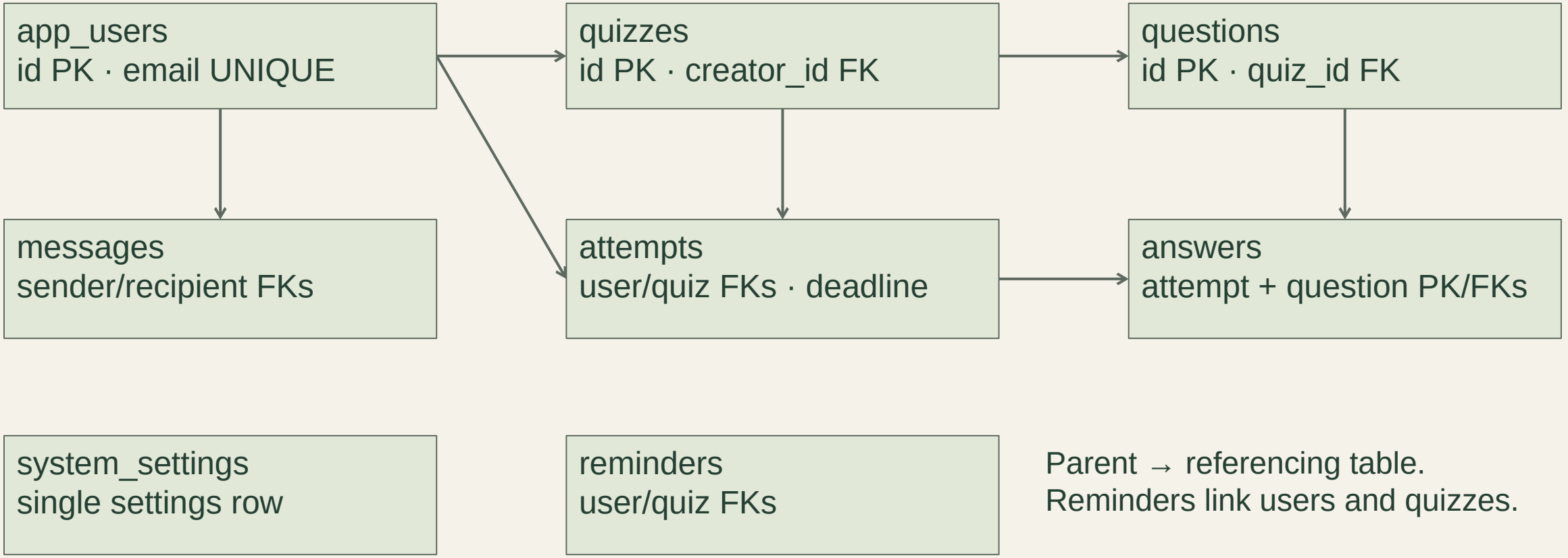
Administrator

Manage accounts and roles, approve content and configure the platform.

Application architecture



Database relationships



Parent → referencing table.
Reminders link users and quizzes.

Prepared JDBC and transaction boundaries

Bound parameters keep submitted values separate from SQL.

```
connection.prepareStatement(sql, new String[] {"id"});
statement.setObject(i + 1, args[i]);

SELECT * FROM attempts WHERE id = ? FOR UPDATE
```

Portable persistence

Generated identities, foreign keys and indexed history queries.

Consistent writes

A transaction and row lock serialize answer saving and submission.

Core Java concepts in working features

Interfaces and polymorphism

ScoringPolicy → MultipleChoiceScoringPolicy calculates quiz results.

Encapsulation and validation

QuizScore record validates score bounds and computes pass status.

Generics and collections

ReportFormatter<T>, queryForObject<T>, List<Map<String, Object>>.

Inheritance and exceptions

PerformanceReportServlet extends HttpServlet; controlled HTTP errors.

Servlet report export

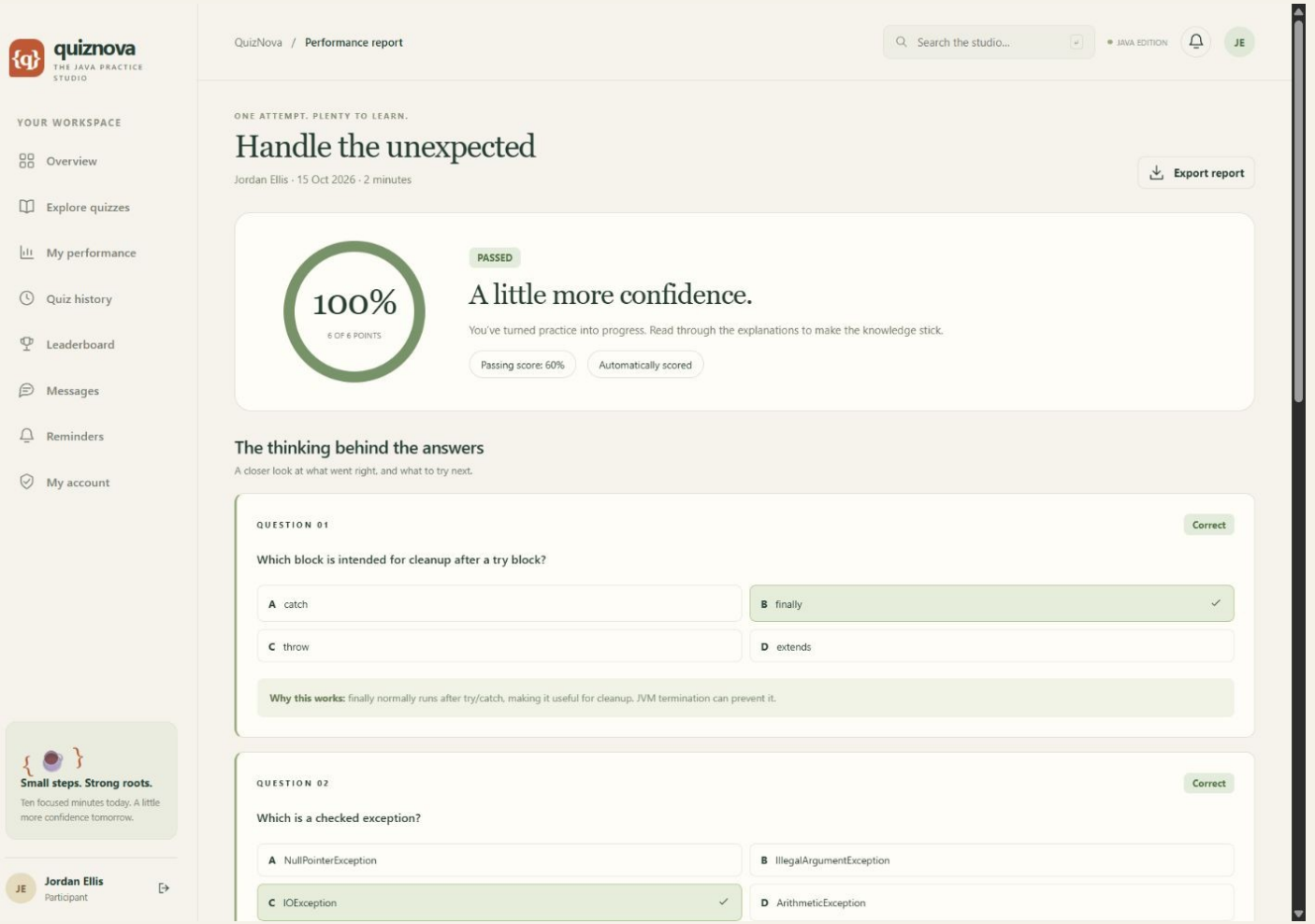
GET /reports/attempts/{id}

Read authenticated principal

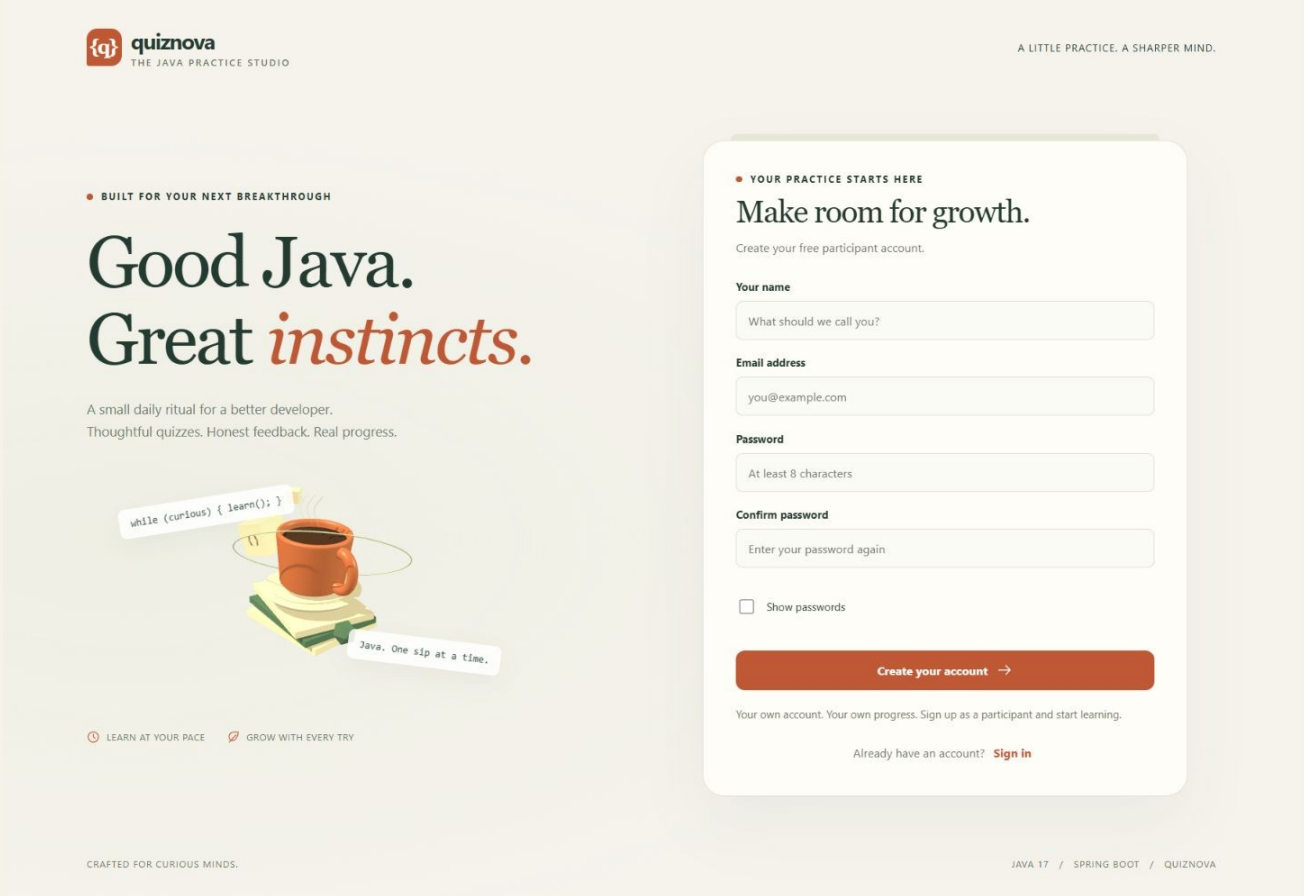
Validate path and ownership

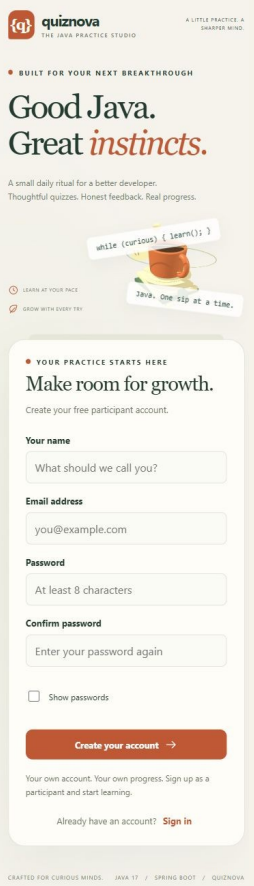
Load a submitted result

Write a UTF-8 attachment



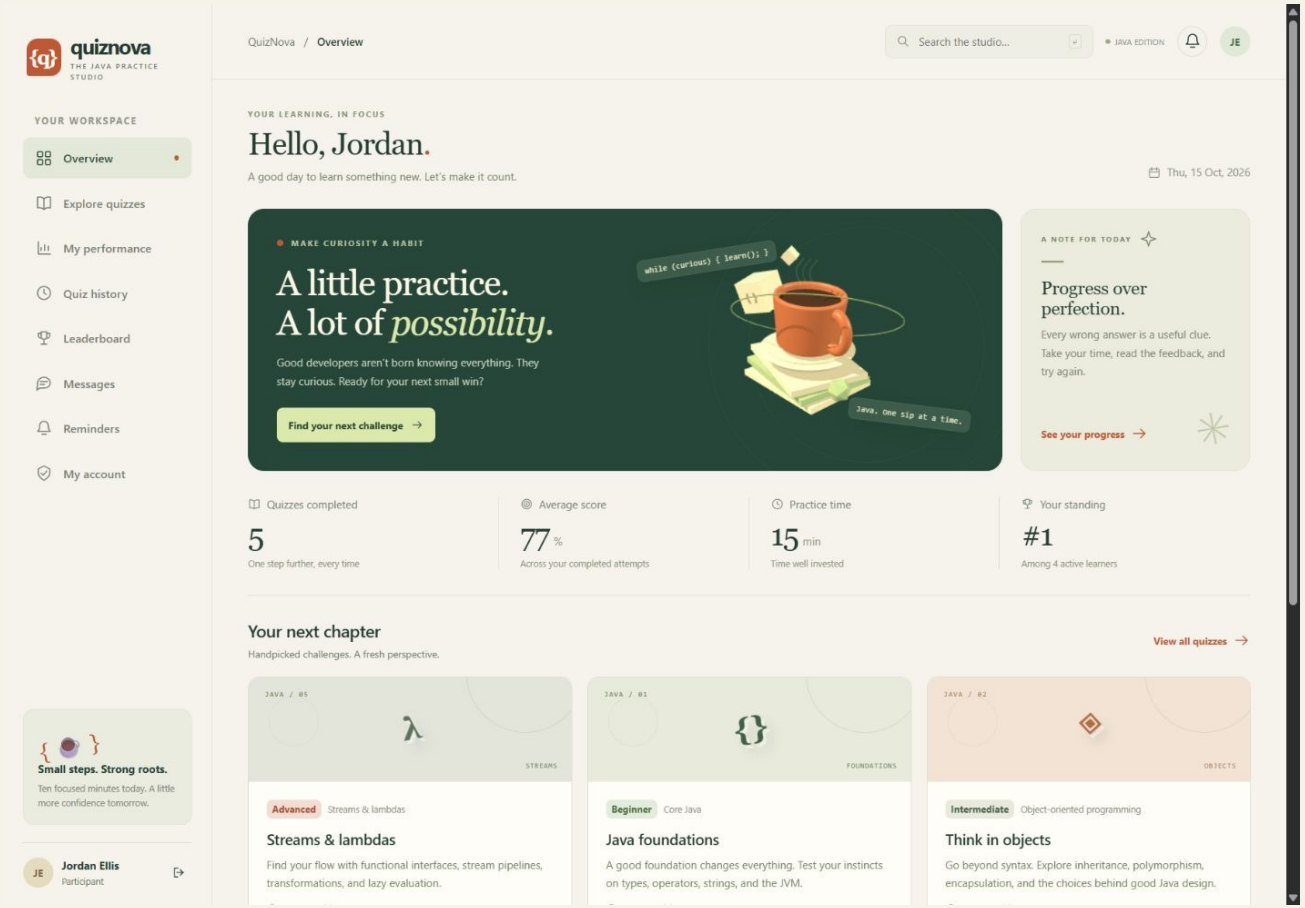
Registration before participation

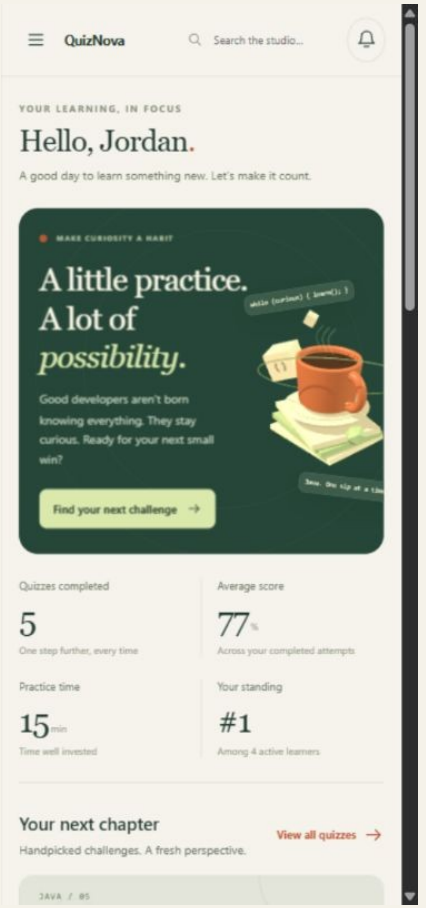




Participant signup → sign in → own workspace

Participant workspace



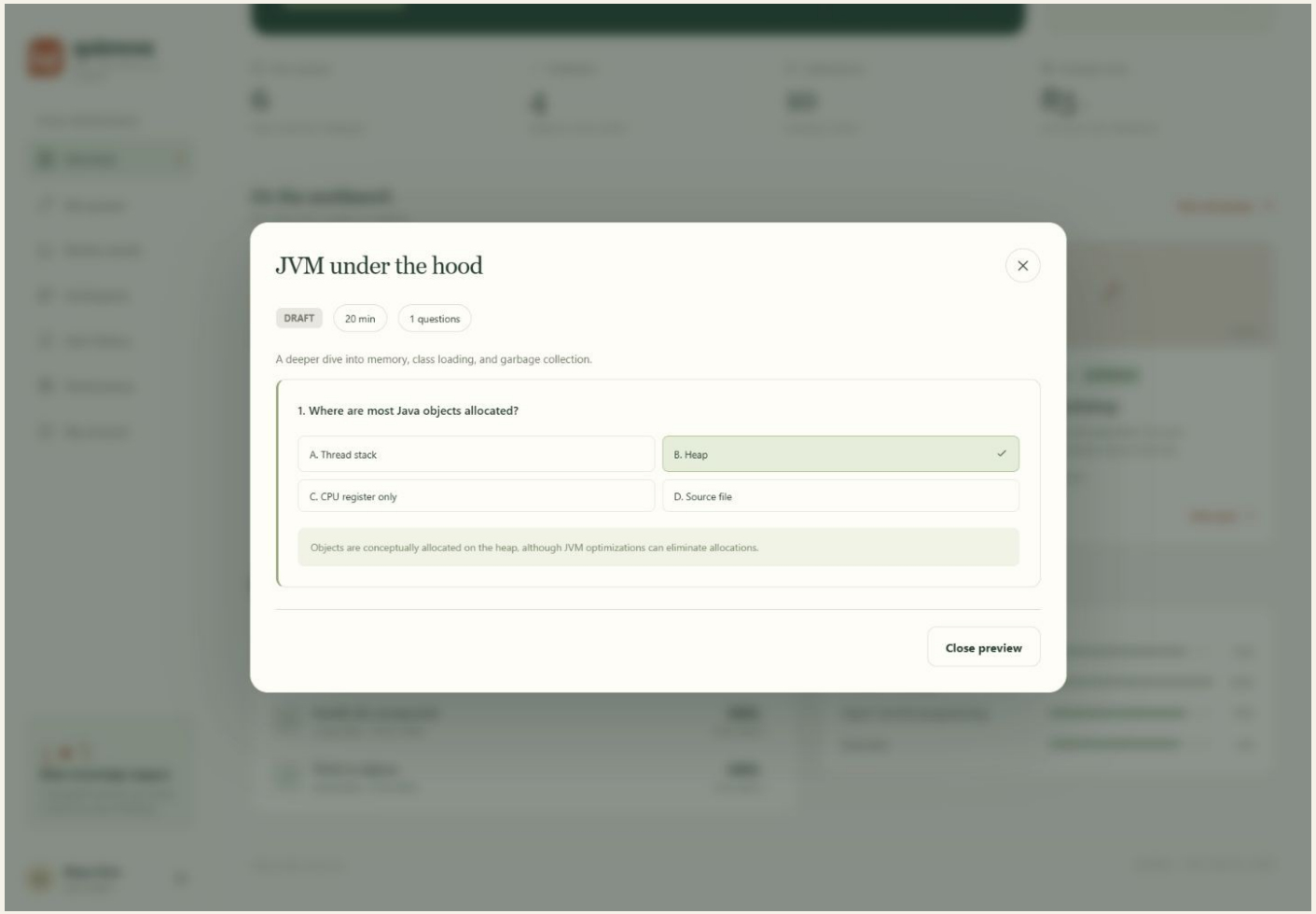


Approved quizzes, recent attempts and personal standing

Creator content and publication workflow

Draft
↓
Pending review
↓
Approved / rejected

Questions include four choices, an answer key and an explanation.



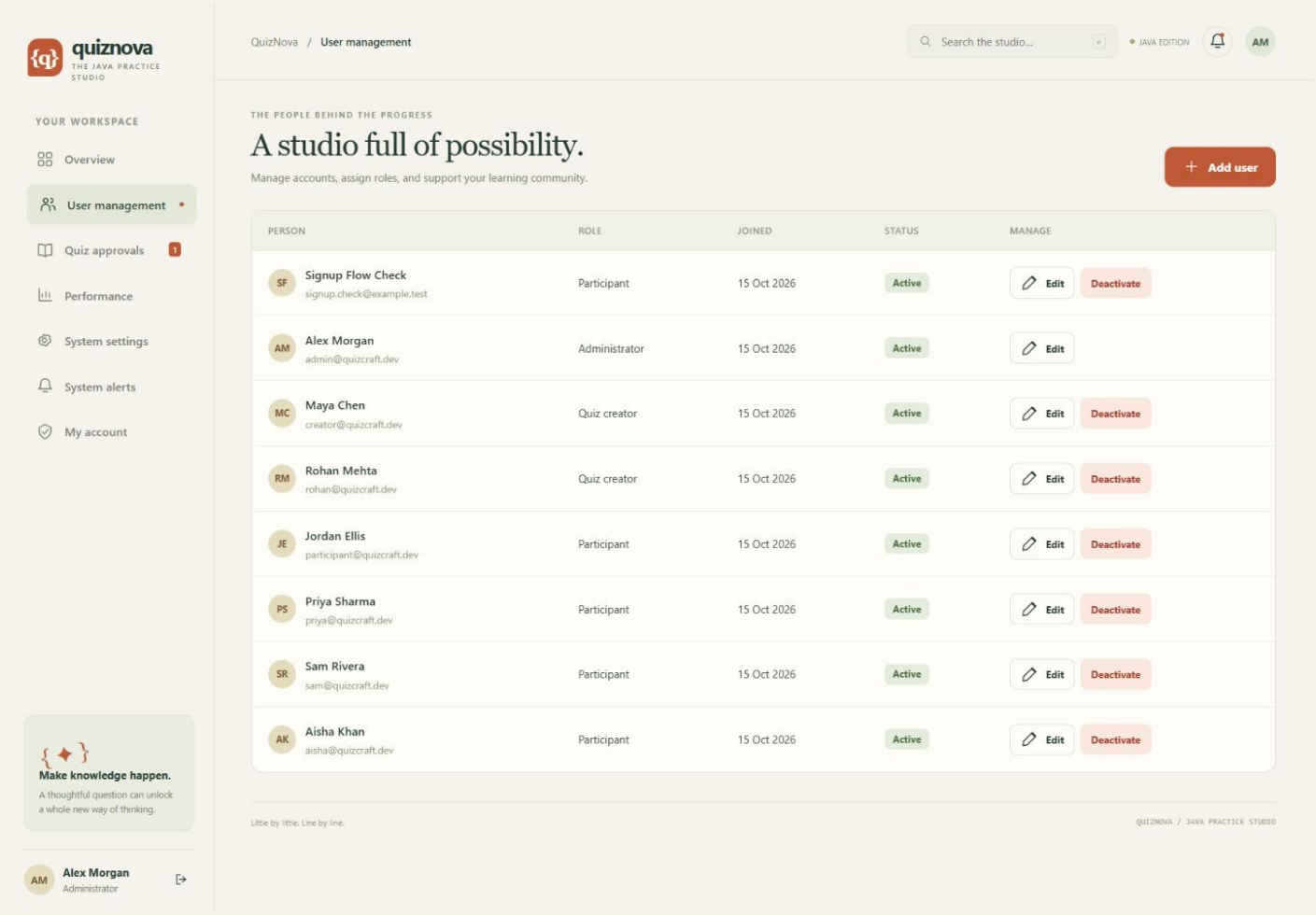
Admin controls

Accounts and roles

Create users, assign creators and deactivate access.

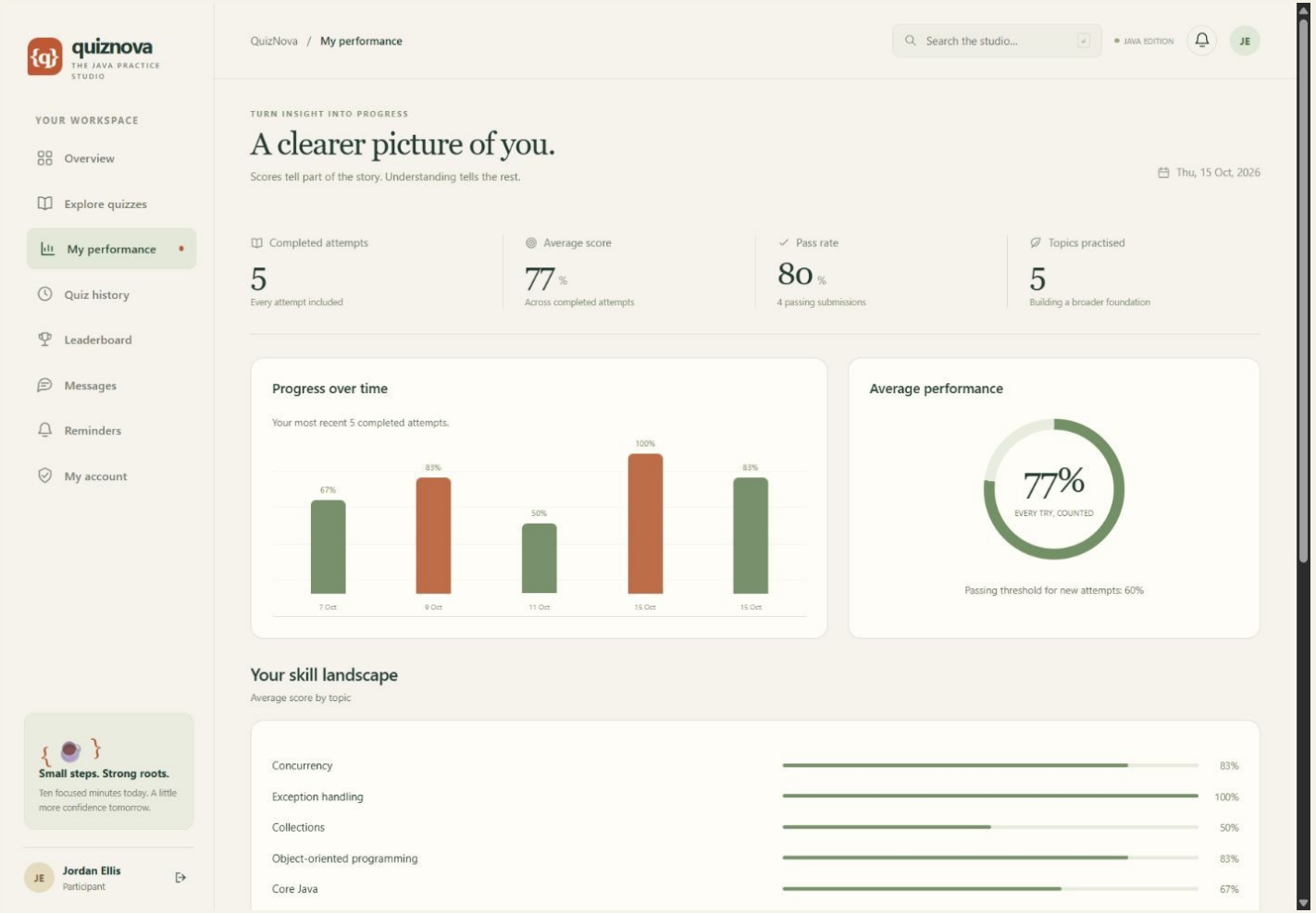
Content and settings

Approve or return quizzes; manage pass and attempt settings.



Performance across attempts

- Scores over time
- Topic averages
- Pass rate
- Answer explanations
- Creator review feedback



Timing, concurrency and verification

21 tests passed

H2 suite and PostgreSQL CI both passed.

Server owns the deadline

Saved answers survive refresh; late choices cannot change a settled result.

Concurrent requests

Six simultaneous starts resolve to one active attempt using a database lock.

Access and lifecycle checks

Role/ownership, scoring, expiry, bootstrap and real servlet HTTP tests.

Java web marking rubric

Criterion	Maximum	Evidence in this project
Problem understanding & design	8	Roles, architecture and database design
Core Java concepts	10	Interfaces, record, generics, exceptions
Database integration (JDBC)	8	Prepared JDBC, constraints, transactions
Servlets & web integration	7	REST routes and native report servlet
Total available	**33**	**Reviewer awards the final score**

This is the web rubric. The separate GUI rubric is a different category.

Run and review the project

Local requirements

Java 17 and Maven 3.9+; run mvn spring-boot:run.

Demo walkthrough

Register → sign in → take quiz → report → creator → admin approval.

Repository documentation

README, architecture, Review 1 evidence, deployment guide and viva notes.

github.com/muskanbuilds11/java-online-quiz-platform

Local demo login credentials

Role	Email address
Administrator	admin@quizcraft.dev
Quiz creator	creator@quizcraft.dev
Participant	participant@quizcraft.dev

Password for all three: JavaLab@123

Normal users create an account first. Admin creates creator accounts.

Cloud deployment

Render Free + Neon PostgreSQL

Java Docker service and a persistent PostgreSQL database in Singapore.

Private administrator access

Use the configured admin email and private password. Local demo logins stay local.

Free hosting behavior

The service sleeps after inactivity. Allow 50 seconds or more for a cold start.

https://quiznova-muskanbuilds11.onrender.com